

The Conversion Engine — Decision Memo

To: Tenacious CEO & CFO From: Charlie Lijam · TRP1 Week 10 Date: 2026-04-25 Repo: The_Conversion_Engine / main

1. EXECUTIVE SUMMARY

The Conversion Engine is a fully automated B2B outreach pipeline that enriches prospects from three public data sources (Crunchbase, layoffs.fyi, careers pages), classifies them into four ICP segments with a confidence-gated LLM email, and syncs every event to HubSpot in real time. On Tenacious-specific failure probes the system cuts bench over-commitment from 7/10 to 2/10 ($p < 0.001$) and tone non-compliance from 61% to 91% ($p = 0.003$), at a cost of \$0.015 per qualified lead — 333× under the \$5 hard cap. We recommend a 30-day Segment 1 pilot at 50 leads/week before any further expansion.

2. COST PER QUALIFIED LEAD

Component	Source	Amount
LLM composition (150 dev-slice simulations)	eval/trace_log.jsonl	\$2.99 total · \$0.0199 avg/sim
Tone guard (Haiku 4.5, ~100 tokens/call)	agent/tone_guard.py	\$0.002/call
Expected regeneration (30% retry rate)	method/ablation_results.json	\$0.0006/lead
Total per qualified lead		\$0.015 (hard cap: \$5)

Rig cost: \$0.00 — all simulations ran on challenge-week provisioned API keys. No Resend or AT charges (kill-switch unset). Source: eval/score_log.json avg_agent_cost=0.0199

3. SPEED-TO-LEAD — STALLED-THREAD RATE

Definition used: A thread is *stalled* when no outbound agent action (email send, SMS, or booking-link delivery) occurs within 4 hours of an inbound prospect event (reply email or inbound SMS webhook). This is measurable directly from data/threads.json timestamps.

Condition	Stall rate (stalls / inbound events)	Measured response latency	Source
Tenacious manual (CEO/CFO estimate)	30–40% stalled >4 h	2–4 h median partner recall delay	Internal CEO/CFO estimate
System — challenge-week threads (n=3)	0 / 3 = 0% stalled	p50: 4.0 s · max: 4.1 s	data/threads.json

Honest caveat: n=3 reply events (all simulated via webhook during challenge week — kill-switch prevented organic prospect replies). A 0% stall rate at n=3 is structurally guaranteed by synchronous dispatch in agent/email_handler.py, not a statistical finding. Confirmation of real-world stall rate reduction requires 30-day pilot with live prospect replies. Projected live stall rate $\leq 20\%$ based on assumption that context-recall delay drives 50% of manual stalls: $40\% \times (1 - 0.50) = 20\%$.

4. COMPETITIVE-GAP OUTBOUND PERFORMANCE

Outbound variant	Challenge-week sends	Signal injected	Challenge-week replies	Benchmark projection
Full method (non-abstained)	10 (staff sink)	100% — hiring_brief + gap_brief in system prompt	3 simulated (organic: 0)	7–12% (Woodpecker 2024)
Baseline (generic pitch)	0 sent this week	0%	—	2–4% (Woodpecker 2024)
Abstained (confidence <0.40)	18% of triggers	Generic only	—	—

Honest caveat: No organic reply rate is measurable — kill-switch routed all 10 challenge-week emails to the staff sink (chalielijam@gmail.com); 3 replies are simulated webhook calls, not prospect responses. The 100% signal-injection rate is structural (code inspection of agent/llm_composer.py: compose_cold_email() always injects hiring_brief + gap_brief). Reply-rate delta between full-method and baseline is a benchmark projection only (Woodpecker 2024 B2B cold email, signal-grounded vs. generic). A minimum $n \approx 200$ organic sends is required to measure a statistically meaningful reply rate at this effect size. Source: agent/icp_classifier.py · data/threads.json · agent/llm_composer.py

5. PILOT SCOPE RECOMMENDATION

Segment: Segment 1 — Recently Funded Series A/B (clearest signal, highest classifier confidence 0.84, lowest offshore-perception risk).
Volume: 50 leads/week × 4 weeks = 200 leads. SDR reviews each draft before send (~2 h/week).
Budget: \$3 LLM cost/month. Kill-switch stays unset until SDR approval gate is wired.
Success criterion: ≥8 qualified discovery calls booked in 30 days (4% meeting rate — conservative below expected 6.4%). If met: expand to Segment 3 and raise volume to 100/week.

6. FOUR FAILURE MODES T²-BENCH DOES NOT CAPTURE

Failure	What it is	Why benchmark misses it	What would catch it	Cost to add
Bench over-commitment	Agent commits "3 Go engineers in 2 weeks" when bench_summary.json is 48–72 h stale and the stack is fully engaged	τ^2 -Bench retail has no bench capacity state; tasks never test capacity-constrained commitments	Probe set P-009–P-012 (already built): inject stale bench file, ask for capacity, check response against actual counts. Needs weekly freshness check.	1 engineer-day to automate bench-staleness injection; \$0.20/probe run
Offshore-perception language	Segment 2 pitch uses "replace higher-cost roles" framing to a CTO who had a failed offshore engagement — triggers immediate complaint or delete	τ^2 -Bench retail users never have offshore-sensitivity backstory; no cultural context injected	Persona-augmented probes: inject CTO persona with "prior failed offshore engagement" flag; score whether agent avoids cost-arbitrage framing. P-013–P-015 partially cover this.	2 engineer-days for persona library; \$0.50/probe run with persona injection
ICP conflict signals (P-004)	Company had layoff 90d ago AND new CTO 45d ago; agent sends email referencing both signals — sounds researched-but-confused	τ^2 -Bench tasks have single clean intents; no overlapping conflicting business signals in state	Multi-signal conflict probes: seed prospect with 2+ overlapping signals, verify agent picks exactly one and hedges the other. Requires HiringSignalBrief with honesty_flags set.	1 engineer-day; \$0.15/probe run
Multi-thread leakage	Two Tenacious SDRs contact CEO and CTO of same company in same week with different segment pitches — prospect receives contradictory framing (\$108K deal risk)	τ^2 -Bench is single-agent single-session; no shared state across parallel agents	Parallel-thread probes: run two agent instances against same company_id in thread_store; verify second agent detects active thread and suppresses outreach. P-016–P-018 cover this.	3 engineer-days for parallel harness; \$0.40/probe run

7. PUBLIC-SIGNAL LOSSINESS — AI MATURITY SCORING

Profile	What the system sees	What the agent does wrong	Business impact
Quietly sophisticated / publicly silent <i>e.g. fintech with proprietary ML stack, no public blog, careers behind login wall</i>	0 AI-adjacent open roles detected, no named ML leadership on LinkedIn, no GitHub OSS activity → ai_maturity_score = 0 , weak_signal = true	Agent abstains or sends generic exploratory opener with no signal grounding. Pitch undersells Tenacious capability. Prospect (who is actually AI-mature) reads it as uninformed outreach and ignores it.	False negative: missed warm lead. At \$36K ACV and 15% close rate: expected \$5,400 lost per false-negative event. Estimated false-negative rate: 12–18% of AI-mature prospects in closed-career-page sectors.
Loud but shallow <i>e.g. Series A startup with "AI-first" in every press release, 0 actual ML roles, no shipped ML product</i>	High executive commentary score, funding event detected, 2–3 AI buzzword job titles → ai_maturity_score = 2 , segment_confidence = 0.78	Agent sends signal-grounded email citing "ML platform gap vs. Tecton" — but prospect has no ML platform. Pitch lands as confused and off-target. CTO forwards it to team as an example of AI vendor noise.	False positive: brand damage. Cost: \$500 reputation estimate per wrong-signal email × 5% false-positive rate on 1,000 emails = \$25,000 expected reputation cost. Trigger for kill-switch Threshold 3 (>3 wrong-signal HubSpot flags per 7-day window).

8. ONE HONEST UNRESOLVED FAILURE — P-004

P-004: Layoff + New CTO (Conflicting Signals) — Residual trigger rate 3/10 after fix

What happens: Prospect company had a layoff 90 days ago (Segment 2 — restructuring signal) and a new CTO joined 45 days ago (Segment 3 — leadership transition signal). The classifier picks Segment 3 (leadership transition wins on recency), but the LLM email references both the layoff and the new CTO in the same sentence. The combined framing reads as "we know you laid people off and got a new boss" — aggressive and off-putting.

Why unresolved: The `honesty_flags` field now includes `conflicting_segment_signals` when both events are present. But the LLM system prompt does not yet have a hard rule suppressing the secondary signal from the email body. Fix requires a prompt constraint: "If `honesty_flags` includes `conflicting_segment_signals`, reference only the primary segment signal in the email body."

Business impact if deployed: Estimated 8% of Segment 2/3 prospects have overlapping signals. At 50 leads/week $\times$ 8% = 4 conflicted leads/week. Each wrong email costs \$500 reputation + \$18,000 expected deal loss ($P(\text{close}) \times \text{ACV}$). **Expected weekly impact unresolved: ~\$72,500.** Do not expand beyond Segment 1 until P-004 is resolved.